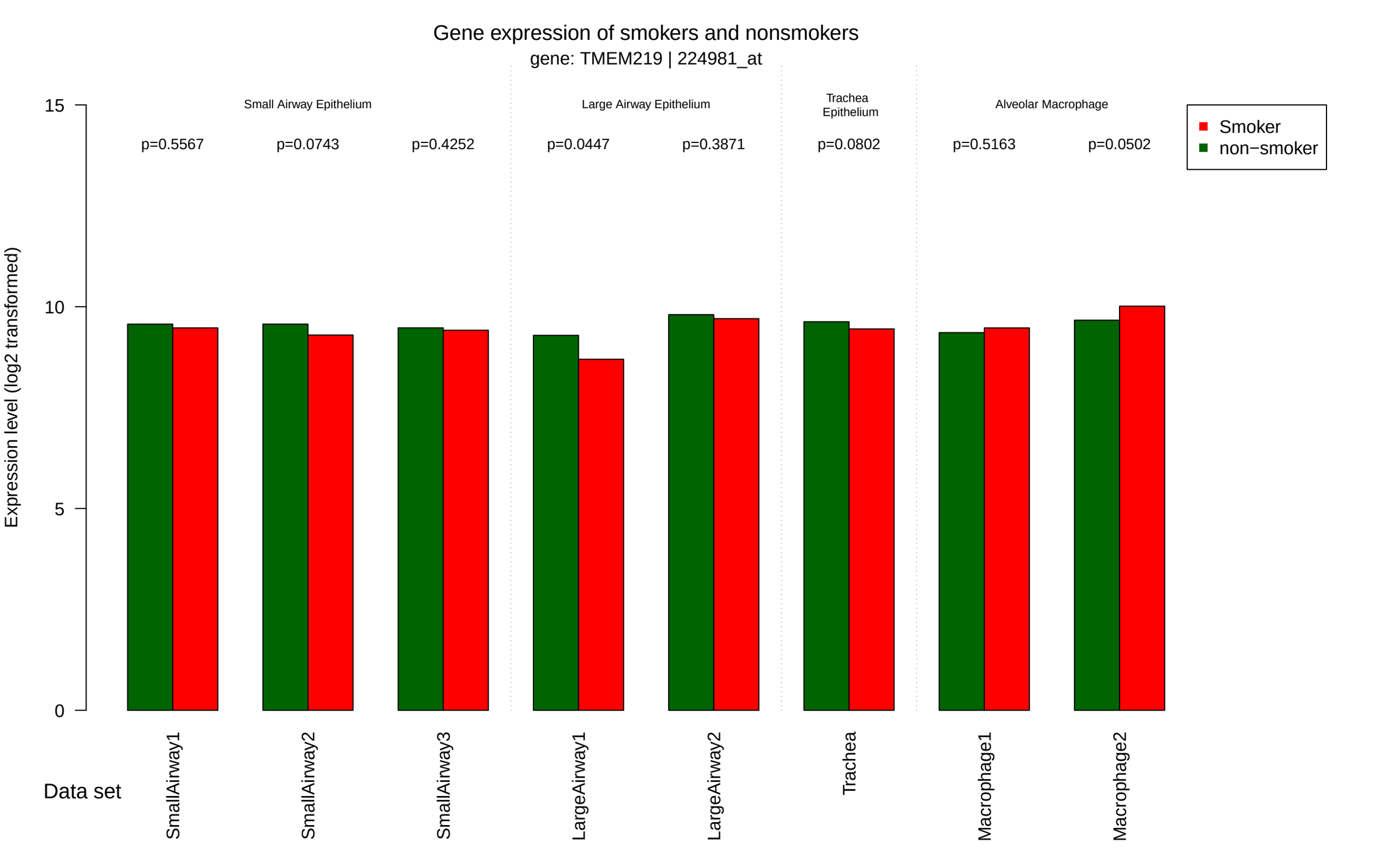
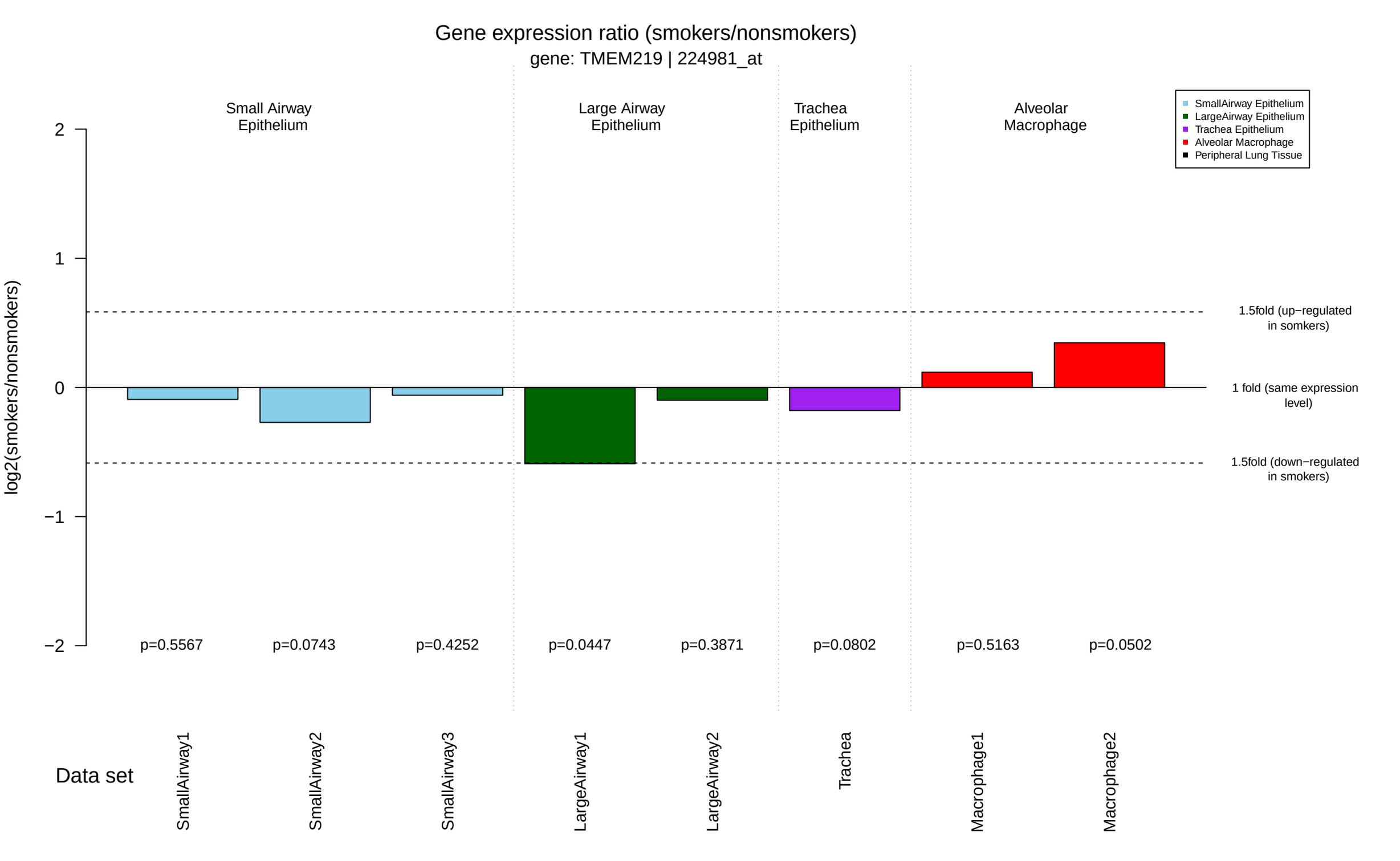
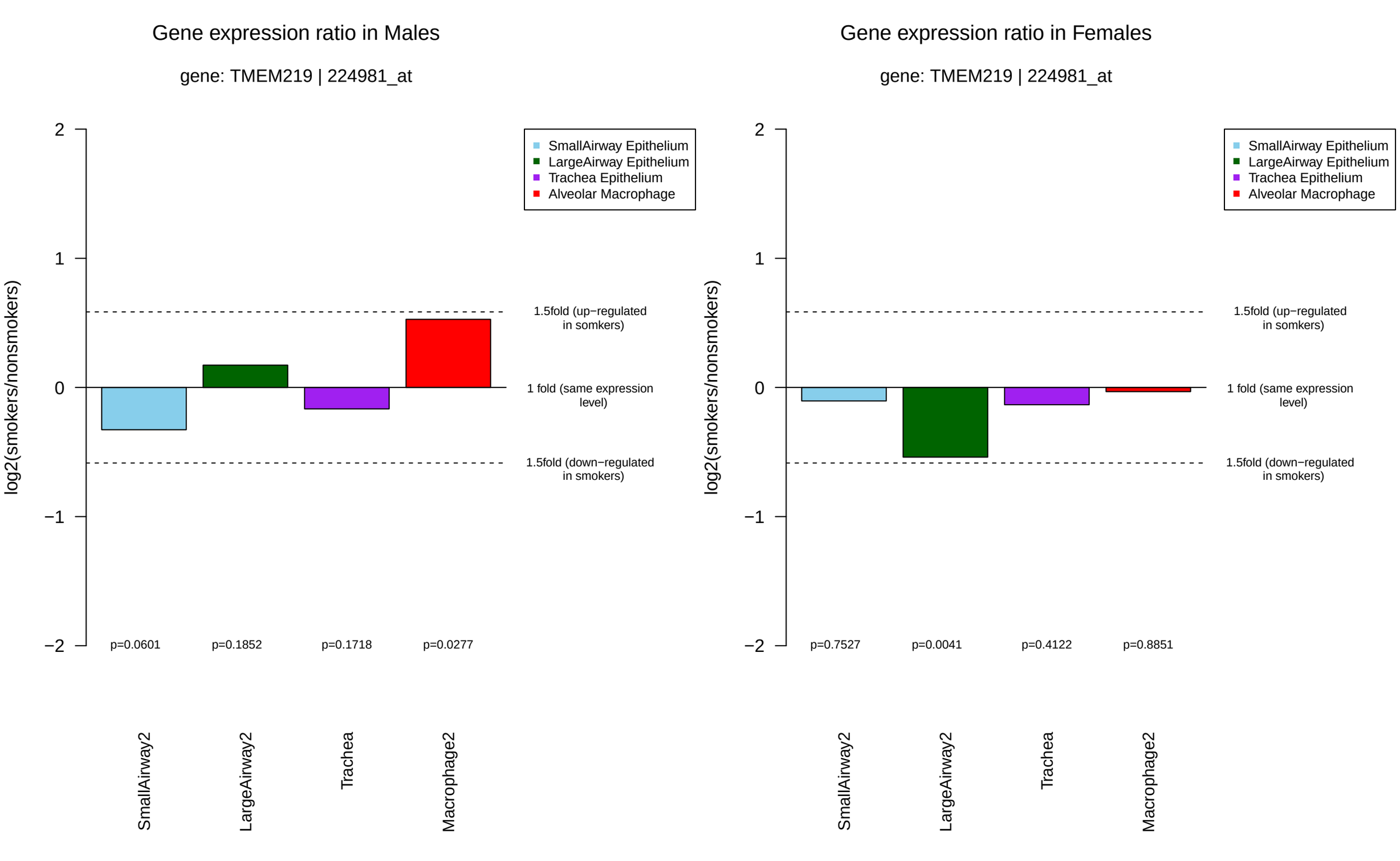
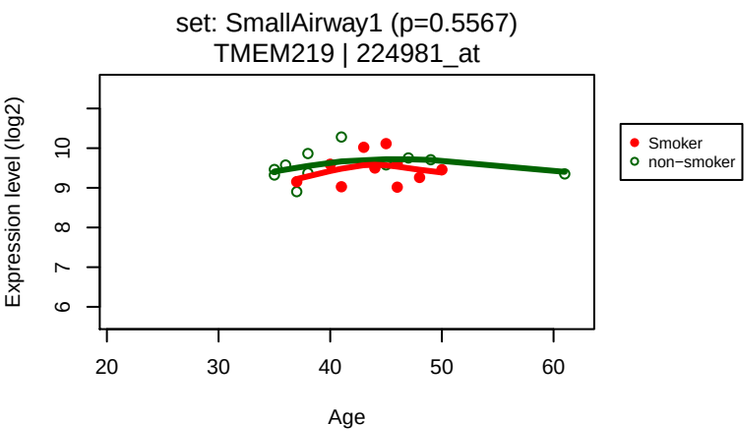
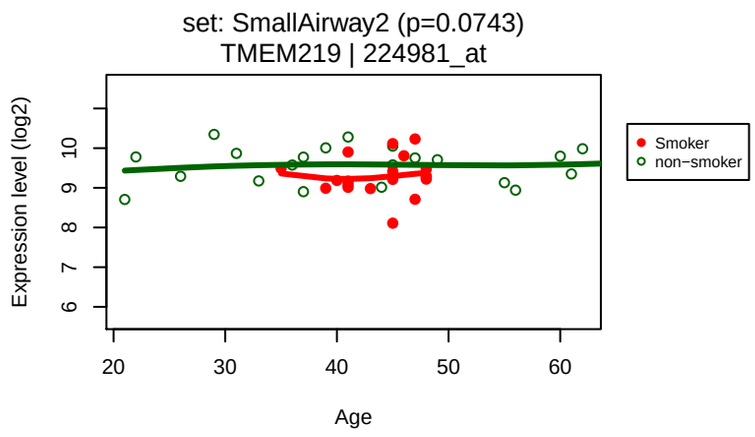
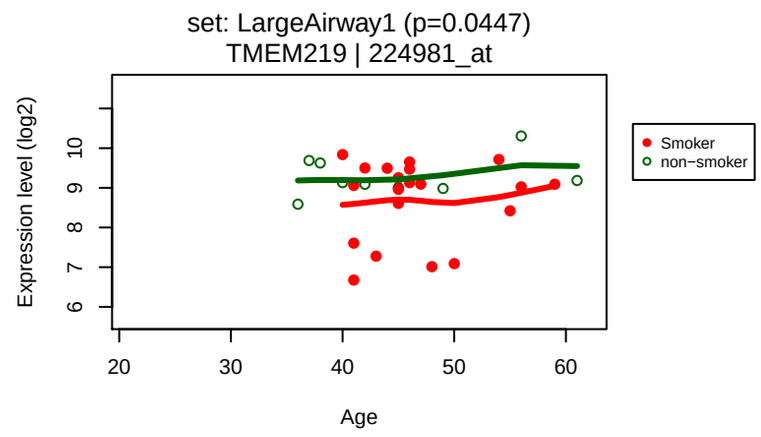

Gene expression of smokers and nonsmokers
gene: TMEM219 | 224981_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5567
p=0.0743
p=0.4252
p=0.0447
p=0.3871
p=0.0802
p=0.5163
p=0.0502
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

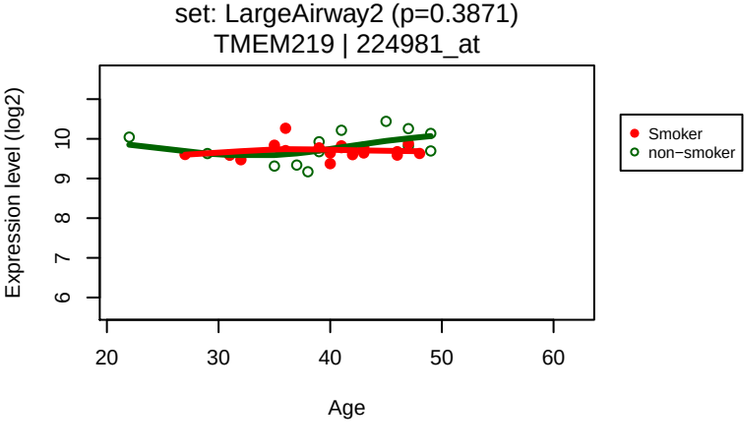

Gene expression ratio (smokers/nonsmokers)
gene: TMEM219 | 224981_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5567
p=0.0743
p=0.4252
p=0.0447
p=0.3871
p=0.0802
p=0.5163
p=0.0502
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

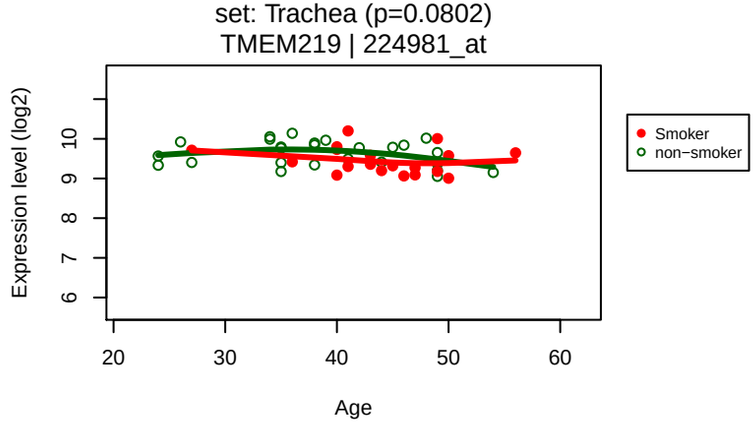

Gene expression ratio in Males
gene: TMEM219 | 224981_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0601
p=0.1852
p=0.1718
p=0.0277
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM219 | 224981_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7527
p=0.0041
p=0.4122
p=0.8851
SmallAirway2
LargeAirway2
Trachea
Macrophage2

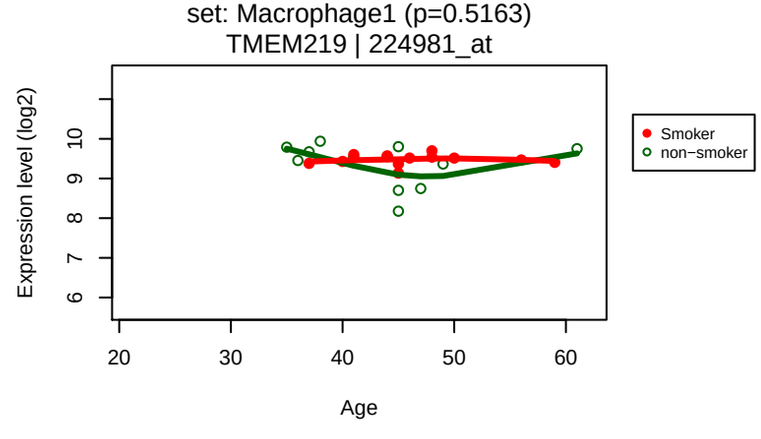

set: SmallAirway1 (p=0.5567)
TMEM219 | 224981_at
Expression level (log2)
Age
Smoker
non−smoker

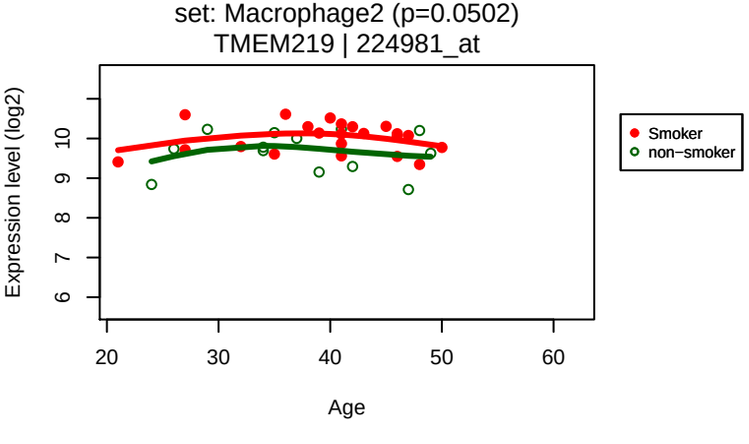

set: SmallAirway2 (p=0.0743)
TMEM219 | 224981_at
Expression level (log2)
Age
Smoker
non−smoker

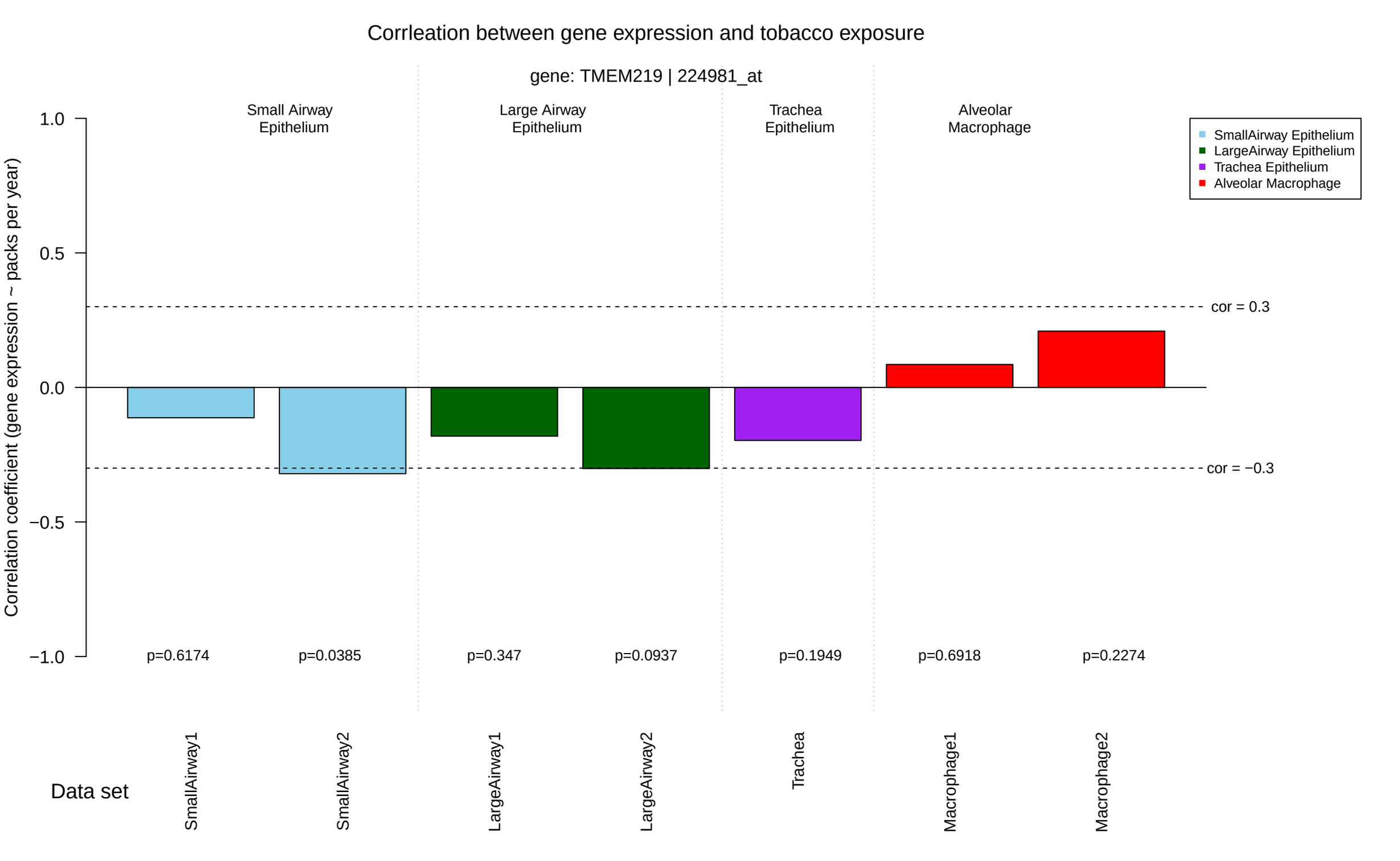

set: LargeAirway1 (p=0.0447)
TMEM219 | 224981_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3871)
TMEM219 | 224981_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0802)
TMEM219 | 224981_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5163)
TMEM219 | 224981_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0502)
TMEM219 | 224981_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM219 | 224981_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6174
p=0.0385
p=0.347
p=0.0937
p=0.1949
p=0.6918
p=0.2274
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2